Class XII Session 2025-26
Subject - Mathematics
Sample Question Paper - 1

Time Allowed: 30 Minutes.

General Instructions:

1. This Question paper contains 38 questions. All questions are compulsory.
2. This Question paper is divided into five Sections - A, .
3. In Section A, Questions no. 1 to 18 are multiple choice questions (MCQs) and Questions no. 19 and 20 are Assertion-Reason based questions of 1 mark each.
4. Use of calculators is not allowed.

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Section A

1. The number of all possible matrices of order 3×3 with each entry 0 or 1 is **[1]**

   a) 18 b) 27

   c) 512 d) 81

2. For non-singular square matrices A and B of the same order, we have adj (AB) = ? **[1]**

   a) (adj B) (adj A) b) |AB|

   c) (adj B).|A| d) (adj A).|B|

3. If $\Delta = \begin{vmatrix} a & h & g \\ h & b & f \\ g & f & c \end{vmatrix}$, then the cofactor A_{21} is **[1]**

   a) hc - fg b) -(hc + fg)

   c) fg - hc d) fg + hc

4. Which of the following statements is true for the function f(x) = $\begin{cases} x^2 + 3, & x \neq 0 \\ 1, & x = 0 \end{cases}$? **[1]**

   a) f(x) is continuous and differentiable $\forall x \in R$ b) f(x) is continuous and differentiable $\forall x \in R$ - {0}

   c) f(x) is discontinuous at infinitely many d) f(x) is continuous $\forall x \in R$

points

5. The angle between the lines $\frac{x}{2} = \frac{y}{2} = \frac{z}{1}$ and $\frac{x-5}{4} = \frac{y-2}{1} = \frac{z-3}{8}$ is **[1]**

a) $\cos^{-1}\left(\frac{3}{4}\right)$ b) $\cos^{-1}\left(\frac{2}{3}\right)$

c) $\frac{\pi}{3}$ d) $\cos^{-1}\left(\frac{5}{6}\right)$

6. The order and degree (if defined) of the differential equation, $\left(\frac{d^2y}{dx^2}\right)^2 + \left(\frac{dy}{dx}\right)^3 = x\sin\left(\frac{dy}{dx}\right)$ respectively are: **[1]**

a) 2, degree not defined b) 2, 2

c) 2, 3 d) 1, 3

7. Maximize Z = 50x + 60y , subject to constraints x +2 y ≤ 50 , x +y ≥ 30, x, y ≥ 0. **[1]**

a) 2500 b) 1600

c) 1547 d) 1525

8. The 2 vectors $j + k$ and $3i - j + 4k$ represents the two sides AB and AC, respectively of a △ABC. The length of the median through A is **[1]**

a) $\frac{\sqrt{48}}{2}$ b) $\frac{\sqrt{34}}{2}$

c) $\sqrt{18}$ d) $\sqrt{22}$

9. $\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \cos x dx$ = ? **[1]**

a) 2 b) -1

c) 1 d) 0

10. If A and B are matrices of same order, then (AB′ – BA′) is a **[1]**

a) skew-symmetric matrix b) unit matrix

c) symmetric matrix d) null matrix

11. The linear programming problem minimize Z = 3x + 2y subject to constraints x + y ≥ 8, 3x + 5y ≤15, x ≥ 0 and y ≥ 0, has **[1]**

a) one solution b) infinitely many solutions

c) no feasible solution d) two solutions

12. If $\vec{a} = (2\hat{i} + 4\hat{j} - k)$ and $\vec{b} = (3\hat{i} - 2\hat{j} + \lambda k)$ be such that $\vec{a} \perp \vec{b}$ then λ =? **[1]**

a) 3 b) -3

c) -2 d) 2

13. If $A = \begin{bmatrix} 1 & k & 3 \\ 3 & k & -2 \\ 2 & 3 & -4 \end{bmatrix}$ is singular then k =? **[1]**

a) $\frac{16}{3}$ b) $\frac{33}{2}$

c) $\frac{34}{5}$ d) $\frac{33}{4}$

14. If $P(A \mid B) = P(A' \mid B)$, then which of the following statements is true? **[1]**

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a) P(A) = 2 P(B)

b) $P(A \cap B) = 2P(B)$

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c) $P(A \cap B) = \frac{1}{2}P(B)$ d) $P(A) = P(A')$

15. A homogeneous equation of the form $\frac{dy}{dx} = h\left(\frac{x}{y}\right)$ can be solved by making the substitution **[1]**

a) $y = vx$ b) $v = yx$

c) $x = v$ d) $x = vy$

16. If $|\vec{a}| = 3$, $|\vec{b}| = 4$ and $|\vec{a} \times \vec{b}| = 6$, then the value of $\vec{a} \cdot \vec{b}$ is **[1]**

a) $6\sqrt{3}$ b) $3\sqrt{3}$

c) 6 d) 12

17. If $\sqrt{x} + \sqrt{y} = \sqrt{a}$ then $\frac{dy}{dx} = ?$ **[1]**

a) $\frac{-\sqrt{x}}{\sqrt{y}}$ b) $\frac{1}{2}\frac{-\sqrt{y}}{\sqrt{x}}$

c) $\frac{1}{2}\frac{\sqrt{y}}{\sqrt{x}}$ d) $\frac{-\sqrt{y}}{\sqrt{x}}$

18. The direction ratios of two lines are a, b, c and (b - c), (c - a), (a - b) respectively. The angle between these lines is **[1]**

a) $\frac{\pi}{3}$ b) $\frac{\pi}{4}$

c) $\frac{3\pi}{4}$ d) $\frac{\pi}{2}$

19. **Assertion (A):** f(x) = tan x - x always increases. **[1]**

Reason (R): Any function y = f(x) is increasing if $\frac{dy}{dx} > 0$.

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

20. **Assertion (A):** The Relation R given by R = {(1, 3), (4, 2), (2, 4), (2, 3), (3, 1) on set A = {1, 2, 3, 2} is symmetric. **[1]**

Reason (R): For symmetric Relation $R = R^{-1}$

a) Both A and R are true and R is the correct explanation of A.

b) Both A and R are true but R is not the correct explanation of A.

c) A is true but R is false.

d) A is false but R is true.

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